



MSCA_DF_10 · Introduction to AI for Financial Applications

Machine Learning in Finance

Programme

One day of the four-day course, online, 09:00–17:00, Poznań time.

Timetable

Time	Block	Papers	In class
09:00–09:20	Opener: prediction vs inference	Mullainathan & Spiess (2017)	slides, claims board
09:20–10:10	A1: from factor models to ML forecasts	Fama & French (2015); Gu, Kelly & Xiu (2020)	notebook 01; write-up and round
10:10–10:20	Break		
10:20–11:20	A2: ML forecasts of returns, and who earns the alpha	Gu, Kelly & Xiu (2020); Avramov, Cheng & Metzker (2023)	notebook 01a; debate
11:20–11:30	Break		
11:30–13:00	B: classification → portfolio → frictions	Krauss, Do & Huck (2017)	notebook 02; you pull the levers
13:00–13:45	Lunch		
13:45–14:45	C: credit and the benchmark protocol	Lessmann et al. (2015); Gunnarsson et al. (2021); Fuster et al. (2022); Zeng et al. (2008) as an exercise	notebook 03 (credit lab)
14:45–15:15	A+: economics inside the loss	Chen, Pelger & Zhu (2024)	slides
15:15–15:25	Break		
15:25–16:05	D: replication and robustness	Harvey, Liu & Zhu (2016); Jensen, Kelly & Pedersen (2023); Kelly, Malamud & Zhou (2024) vs Nagel (2025)	notebook 06
16:05–16:55	E: replication from the author's side	Będowska-Sójka, Wójcik & Pele (2025)	notebook 05 (shared screen); robustness round
16:55–17:00	Wrap: seven moves		one-line takeaways

Extra, not in the timetable above: block 07, Cakici, Shahzad, Będowska-Sójka & Zaremba (2024) on machine learning and the cross-section of cryptocurrency returns, with notebook 07_cakici_crypto_ml. It takes Block E from one coin universe to the cross-section. The deck and the notebook are in the folder.

How we work

- **Predict first.** Before we show a result, we ask you to guess it in the Zoom chat: a question with lettered options. Type your letter when we say “go”, not before, so nobody anchors on the others.
- **One seminar group.** Three times during the day you get about three minutes to write your answer on the shared board, and then each of us says one sentence. No breakout rooms.
- **Two rotating roles.** In every block one of you is the **skeptic**, who names one flaw in the paper or in our code, and one is the **runner**, who runs the notebook and reads out the numbers. The roles move around, so everyone does both.
- **One debate.** Gu, Kelly & Xiu against Avramov, Cheng & Metzker: half of us take each side, three minutes to prepare, one minute to make the case.
- **The shared board** is a document with one page per activity; its link is shared on the day.
- **Notebooks:** you run the live notebooks on your own computer if your setup check passed; otherwise you follow the shared screen. Each live notebook runs in under two minutes.

After the day (homework)

Pick one:

	Task	Block
A	A 1–2 page deconstruction note on one core paper, with the hypothesis stated	any
B	A half-page Lessmann-style benchmark protocol for your own data set	C
C	Rerun <code>06_kmz_nagel</code> with training windows of 60 and 120 months: does the momentum reading survive?	D
D	A sketch (bullets or code stub): add a finance penalty inside training, starting from <code>04_chen_pelger_zhu_sketch</code>	A+

Optional notebooks: `01b_gu_fama_french` (Fama–French characteristics in the Gu pipeline) and `03_invoice_timing_simplified` (the invoice-to-cash case, with a leakage hunt).

The slides are in `slides/`, one deck per block.